

SQL Security Analysis

Project Description

Used SQL to analyze login activity and employee data in support of security investigations and targeted system updates. Applied SQL query logic to efficiently filter and extract relevant data.

```
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 35
Server version: 10.5.29-MariaDB-0+deb11u1 Debian 11

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [organization]> clear
MariaDB [organization]> SELECT * FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0 |
| 18 | pwashing | 2022-05-11 | 19:28:50 | US | 192.168.66.142 | 0 |
| 20 | tshah | 2022-05-12 | 18:56:36 | MEXICO | 192.168.109.50 | 0 |
```

Query Logic

By applying **WHERE** conditions, I filtered results based on time, date, location, and department. I combined conditions using logical operators (**AND**, **OR**, **NOT**) to narrow or expand results, and used **LIKE** with **%** to match patterns in text fields such as geographic locations and office names.

Retrieve and Filter Security-Relevant Data

- Identified failed login attempts after business hours to detect potential unauthorized access:

```
SELECT * FROM log_in_attempts
WHERE login_time > '18:00' AND success = FALSE;
```

- Identified login attempts originating outside expected geographic region to detect unusual access patterns:

```
SELECT * FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

- Filtered login activity across specific dates to investigate an incident window:

```
SELECT * FROM log_in_attempts
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

```
MariaDB [organization]> SELECT * FROM employees
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134

- Filtered employee data to pull `device_id` and support targeted system updates based on department and location:

```
SELECT * FROM employees
WHERE department = 'Finance' OR department = 'Sales';
```

```
SELECT * FROM employees
WHERE department = 'Marketing' AND office LIKE 'East%';
```

```
SELECT * FROM employees
WHERE NOT department = 'Information Technology';
```

Summary

I applied filters to SQL queries to get specific information on login attempts and employee machines. I used two different tables, `log_in_attempts` and `employees`. I used the **AND**, **OR**, and **NOT** operators to filter for the specific information needed for each task. I also used **LIKE** and the percentage sign (%) wildcard to filter for patterns. These queries were to enable efficient identification of suspicious activity and support targeted security actions.